

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: Deep Learning
FACULTY : Dr.Arul Raj A.M

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I **Yoga Madha A-192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: Yoga Madha A

Analytical Problem Solving

1. Learning Algorithms (Optimization Behavior)

A machine learning model is trained using gradient descent on a convex loss function. Initially, the learning rate is set very high, and later it is gradually decreased.

Question:

Analyze how the choice of a high initial learning rate followed by decay affects convergence. Under what conditions can this strategy outperform a constant learning rate? Discuss possible risks and justify with reasoning.

Solution:

A **high initial learning rate** allows Gradient Descent to make large updates to the model parameters during the early stages of training. This helps the model reduce the loss quickly and reach the region of the global minimum in fewer iterations. As the learning rate is **gradually decreased (learning rate decay)**, the updates become smaller and more precise, leading to stable convergence and improved accuracy.

This strategy can outperform a **constant learning rate** when:

- The loss function is **convex**, ensuring convergence to the global minimum.
- The dataset is **large**, where faster initial learning reduces training time.
- A **proper learning rate decay schedule** is used, balancing speed and stability.

Possible Risks

- If the initial learning rate is **too high**, the optimizer may **overshoot** the minimum, causing oscillations or divergence.
- If the learning rate **decays too quickly**, learning may slow down before reaching the optimum.
- If the learning rate **decays too slowly**, the model may continue oscillating around the minimum and fail to converge accurately.

Justification

Using a **high initial learning rate followed by gradual decay** combines the advantages of fast learning in the beginning and precise optimization in the later stages. Compared to a constant learning rate, this approach often achieves **faster convergence, better stability, and higher final accuracy**, provided the initial learning rate and decay schedule are chosen appropriately.

2. Maximum Likelihood Estimation (MLE)

Suppose you are given a dataset generated from a Gaussian distribution with unknown mean μ and known variance σ^2

Question:

Derive the Maximum Likelihood Estimator for μ , and analytically explain how the estimate changes when:

- The dataset contains outliers
 - The sample size increases
- What does this imply about the robustness of MLE?

Solution:

For a Gaussian distribution with **unknown mean (μ)** and **known variance (σ^2)**, the Maximum Likelihood Estimator (MLE) of the mean is the **sample mean**:

$$\hat{\mu}^{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i$$

Effect of Outliers

Since the MLE estimate is the **sample mean**, it is highly sensitive to outliers. A few extremely large or small values can significantly shift the estimated mean away from the true value, reducing estimation accuracy.

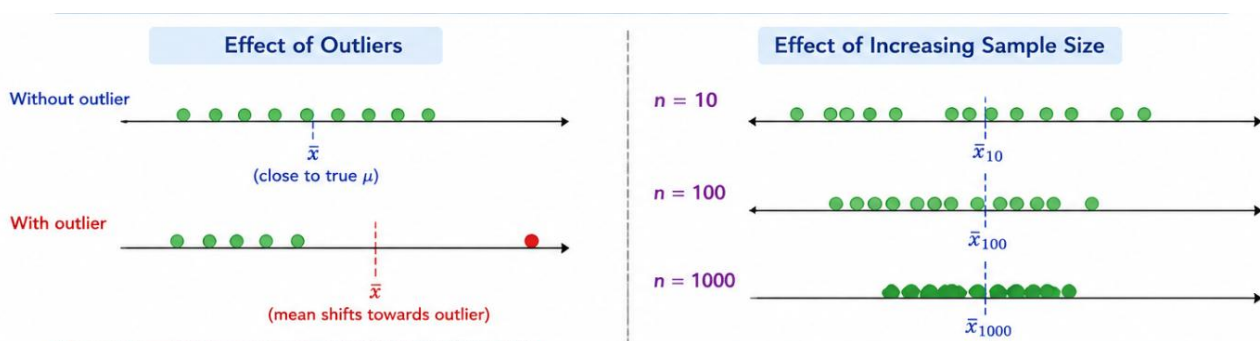
Effect of Increasing Sample Size

As the sample size increases, the MLE estimate becomes more accurate and stable. According to the **Law of Large Numbers**, the sample mean approaches the true population mean. Larger datasets reduce the influence of random variation and improve the reliability of the estimate.

Implication for Robustness of MLE

- **Advantages:**
 - Provides an efficient and unbiased estimate for normally distributed data.
 - Accuracy improves as the sample size increases.
 - Converges to the true parameter with more observations.
- **Limitations:**
 - Not robust to outliers because extreme values directly affect the sample mean.
 - Performs poorly when the data contains significant noise or deviations from the assumed Gaussian distribution.

The MLE for the Gaussian mean is the **sample mean**, which is simple and effective for clean, normally distributed data. However, it is **sensitive to outliers**, making it less robust. As the sample size increases, the estimate becomes more reliable and converges toward the true mean, improving overall estimation accuracy.



3. Building a Machine Learning Algorithm (Model Selection)

You are designing a classification algorithm for a dataset with 10,000 features but only 500 training samples.

Question:

Analyze the challenges involved in training such a model. Propose a strategy (algorithm pipeline) to handle this situation and justify each step (e.g., feature selection, regularization, dimensionality reduction). Explain how your approach mitigates overfitting.

Solution:

A dataset with **10,000 features** and only **500 training samples** presents a **high-dimensional, low-sample-size** problem. The main challenge is **overfitting**, where the model learns noise instead of meaningful patterns. Other challenges include increased computational cost, irrelevant or redundant features, and poor generalization to new data.

Proposed Algorithm Pipeline

1. **Data Preprocessing**
 - Handle missing values and normalize the data.
 - Ensures all features are on a similar scale.
2. **Feature Selection**
 - Remove irrelevant and redundant features using methods such as correlation analysis or L1 (Lasso) feature selection.
 - Reduces the number of features and improves model efficiency.
3. **Dimensionality Reduction**
 - Apply **Principal Component Analysis (PCA)** to reduce feature dimensions while preserving most of the important information.
 - Decreases computational complexity and removes noise.
4. **Model Training with Regularization**
 - Train a classifier such as **Logistic Regression** or **Support Vector Machine (SVM)** with **L2 regularization**.
 - Regularization prevents the model from assigning excessive importance to any single feature.
5. **Cross-Validation**
 - Use **k-fold cross-validation** to evaluate model performance and select the best hyperparameters.
 - Ensures reliable performance on unseen data.

How the Approach Mitigates Overfitting

- **Feature selection** removes unnecessary features that may introduce noise.
- **Dimensionality reduction** compresses the data while retaining important information.
- **Regularization** limits model complexity and prevents large parameter values.
- **Cross-validation** helps detect overfitting and improves model generalization.

4. Neural Networks (Multilayer Perceptron - MLP)

Consider a Multilayer Perceptron with one hidden layer using sigmoid activation functions.

Question:

Explain analytically why adding more hidden neurons increases the representational power of the network. Then, reason why this does not always lead to better performance. Support your answer using concepts like overfitting and generalization.

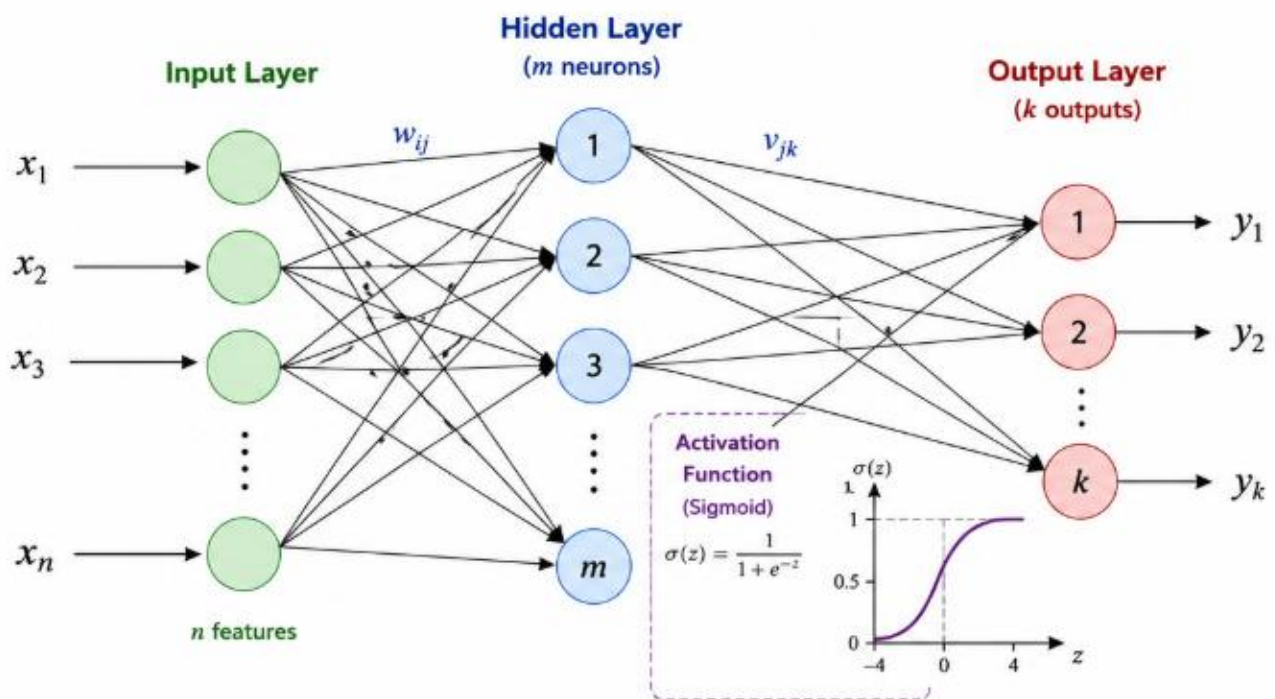
Solution:

A **Multilayer Perceptron (MLP)** with one hidden layer uses **sigmoid activation functions** to learn complex relationships between inputs and outputs. Adding more hidden neurons increases the network's ability to represent complex patterns because each neuron learns different features of the input data. With more neurons, the network can model more complicated decision boundaries and nonlinear relationships.

However, increasing the number of hidden neurons **does not always improve performance**. A larger network contains more parameters, making it more likely to **overfit** the training data. Overfitting occurs when the model learns noise and specific details of the training data instead of the underlying patterns, resulting in poor performance on unseen data.

In contrast, a model that **generalizes well** captures the important patterns in the data and performs accurately on both training and testing datasets. Therefore, simply increasing the number of hidden neurons can improve representational power, but without sufficient training data or proper regularization, it may reduce generalization.

Adding more hidden neurons increases the representational power of an MLP by enabling it to learn more complex nonlinear patterns. However, excessive neurons can lead to **overfitting**, increased computational cost, and reduced generalization. Therefore, the optimal number of hidden neurons should be chosen carefully to balance **model complexity and prediction accuracy**.



5. Backpropagation, SGD & Curse of Dimensionality

A deep neural network is trained using Stochastic Gradient Descent (SGD) on a very high-dimensional dataset.

Question:

Analyze how the **curse of dimensionality** impacts:

- Gradient updates in backpropagation
- Convergence speed of SGD
- Model generalization

Propose at least two techniques to overcome these issues and justify them analytically.

Solution:

When a deep neural network is trained on a **high-dimensional dataset**, the **curse of dimensionality** makes learning more difficult because the number of features is very large compared to the available data.

Impact on Gradient Updates in Backpropagation

- High-dimensional data increases the number of model parameters.
- Gradients become noisy and unstable, making weight updates less effective.
- This can lead to **vanishing or exploding gradients**, slowing the learning process.

Impact on Convergence Speed of SGD

- SGD requires more iterations to find the optimal solution.
- The optimization process becomes slower because the parameter space is much larger.
- Training becomes computationally expensive and convergence may be unstable.

Impact on Model Generalization

- High-dimensional models are more likely to **overfit** the training data.
- The model learns noise instead of meaningful patterns, resulting in poor performance on unseen data.

Techniques to Overcome These Issues

1. Dimensionality Reduction (PCA or Feature Selection)

- Reduces the number of input features by removing redundant or less important information.
- This decreases model complexity, speeds up training, and improves generalization.

2. Regularization (L1/L2 or Dropout)

- Regularization limits the complexity of the model by preventing excessively large weights.
- Dropout randomly deactivates neurons during training, reducing overfitting and improving robustness.

Justification

Dimensionality reduction simplifies the input space, making gradient updates more stable and SGD faster. Regularization prevents the model from memorizing the training data, leading to better generalization. Together, these techniques reduce overfitting, improve convergence speed, and produce a more accurate and reliable deep learning model.